

Technology News

'Evolving' robots

Taking cue from biological evolution, the scientists have developed a robot with a software 'brain' that automatically grows in size and complexity as its physical body develops. The only but major difference is that it does this trick in hours.

Existing robots cannot usually cope with physical changes—the addition of a sensor or new type of limb, say—without a complete redesign of their control software, which can be time-consuming and expensive. So artificial intelligence engineer Christopher MacLeod and his colleagues at the Robert Gordon University in Aberdeen, UK, created a robot that adapts to such changes by mimicking biological evolution. "If we want to make really complex humanoid robots with ever more sensors and more complex behaviours, it is critical that they are able to grow in complexity over time—just like biological creatures did," he says.

The brain of MacLeod's robot assigns new clusters of 'neurons' to adapt to new additions to its body. The robot is controlled by a neural network—software that mimics the brain's learning process. This comprises a set of interconnected processing nodes which can be trained to produce desired actions. Such actions are shaped by adjusting the importance, or weighting, of the input signals to each node. Certain combinations of these sensor inputs cause the node to fire a signal—to drive a motor, for example. If this action works, the combination is kept. If it fails, and the robot falls over, the robot will make adjustments and try something different next time.

Flexible electronic books

Screen breakage is a common complaint about today's electronic books. However, UK-based Plastic Logic



World's first flexible electronic book from Plastic Logic

claims to have developed the world's first flexible electronic book with a bendy screen that can withstand a lot of tumble.

Currently, the two most popular e-books on the market, the Sony Reader and Amazon Kindle, are paperback book-sized devices that use B&W electronic ink displays. These consist of a plastic sheet containing pixel-sized voids, each filled with B&W ink particles. Electric fields attract the ink to the top of these voids to display print. The problem is, the transistors that apply these electric fields sit on a layer of glass, making the displays fragile.

Plastic Logic has perfected a way of printing polymer transistors onto a layer of bendy plastic, allowing the screens to flex and bounce. This device will have wireless Internet connection and a touch screen, allowing use of a virtual keyboard for annotating text.

Optical clocks are the future

A new generation of clocks, called optical clocks, can keep time without missing a beat in almost two billion years. These clocks are so precise that they will allow satellites to track moving objects to within less than a metre—a

prerequisite for automated cars.

The new devices are known as optical clocks because lasers measure the frequency with which electrons in atoms vibrate. This enables them to divide time into even tinier increments.

The most advanced clock, created by researchers at the National Institute of Standards and Technology (NIST) in Boulder, Colorado, measures the vibrations of electrons in mercury ions and can go 1.7 billion years without missing a beat. A normal wristwatch, by contrast, loses about a second every week.

The most exciting developments are likely to come in the field of global positioning systems (GPS), which track planes, ships and cars. GPS devices receive microwave signals sent by satellites and, by measuring the time these take to arrive, can pinpoint the location of an object on Earth to within 10 metres.

Universal charger for mobile phones

We all need to buy a charger with every new mobile phone. What if you can charge your phone from any available charger? This would not only make life easier for mobile phone